INTELLIGENT FAULT DETECTION AND PREDICTIVE MAINTENANCE FRAMEWORK FOR AUTONOMOUS SPACECRAFT OPERATIONS

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WHY AUTONOMOUS HEALTH MONITORING?

- Space missions like Mangalyaan, Chandrayaan, and Aditya-L1 cannot rely on ground support during critical phases
- Signal round-trip delays make human intervention too slow to prevent failures
- A single undetected engine fault can result in complete mission loss
- Onboard AI systems must monitor, predict, and respond without human input

NASA C-MAPSS-1 — ENGINE TELEMETRY DATASET

- Simulates run-to-failure cycles for 100 turbofan engines
- Each engine has 21 sensor channels recorded every operational cycle
- After removing constant-variance sensors, 14 features retained
- Engine labelled degraded when fewer than 30 cycles remain
- Data normalised to 0-1 range using Min-Max scalin

NASA C-MAPSS-1 TURBOFAN ENGINE DEGRADATION DATASET

FOUR-COMPONENT AI PIPELINE

- Component 1: Isolation Forest — tree-based unsupervised anomaly scorer
- Component 2: Autoencoder Neural Network — detects deviation from healthy reconstruction
- Component 3: Stacked LSTM Classifier — learns degradation patterns across 30-cycle windows
- Component 4: LSTM Regression Model — predicts exact remaining engine life (capped at 125 cycles)

PERFORMANCE METRICS

- Isolation Forest: F1 = 0.69 | Precision = 0.71 | Recall = 0.68
- Autoencoder: F1 = 0.76 | Precision = 0.78 | Recall = 0.74
- LSTM Classifier: F1 = 0.83 | Precision = 0.85 | Recall = 0.82
- RUL Regression MAE = 18.4 cycles | RMSE = 24.7 cycles
- 13 engines in critical zone, 35 in warning, 52 operating normally

RISK-BASED AUTONOMOUS ACTION SYSTEM

- Each engine assigned a composite risk score every cycle
- Formula: 40% from anomaly model agreement + 60% from RUL proximity to failure
- Five response levels: Normal, Monitor, Warning, High Risk, Emergency
- No human approval required — system acts immediately on its own assessment
- Designed for scenarios where ground communication is unavailable

REAL-TIME TELEMETRY DASHBOARD

- Built using Matplotlib FuncAnimation inside Kaggle notebook
- Streams actual sensor readings cycle-by-cycle using trained model scores
- Anomaly markers appear instantly when any model raises a flag
- Separate panels show risk timeline, RUL gauge, and scrolling alert log
- All visualisations driven by real model outputs, not pre-recorded data

SUMMARY AND NEXT STEPS

- Successfully built a three-model ensemble with LSTM achieving F1 of 0.83
- RUL prediction error of 18 cycles is competitive with published literature
- System makes autonomous decisions across five risk categories without human input
- Future: deploy on TensorFlow Lite for onboard edge hardware
- Future: extend to FD002-FD004 with multi-condition domain adaptation
- Future: integrate SHAP explainability for post-mission decision auditing
- Future: apply pipeline to Aditya-L1 solar wind sensor data

THANK YOU